

Project: ChakulaKonnnect – Digital Platform for Affordable Nutritious Food Access

Cybersecurity Deliverable: MVP Security Requirements Checklist

Objective: Ensure the platform is secure, protects user data, and aligns with NDPR and industry best practices.

1. Authentication & Account Security

- User passwords must never be stored in plain text.
 - o Use secure hashing (bcrypt/argon2).
- Secure login/session handling using JWT or server sessions.
- Apply brute-force protection through login rate limiting and lockouts.
- Enforce basic password strength requirements.

2. Role-Based Access Control (Authorization)

- Implement clear user roles: Consumers, Farmers/Vendors, Admin.
- Users must only access their own data (orders, profiles, listings).
- Prevent unauthorized access vulnerabilities such as IDOR.
- Restrict admin privileges and monitor administrative actions.

3. API & Application Security (OWASP Controls)

- Validate and sanitize all inputs to prevent:
 - o SQL Injection
 - o Cross-Site Scripting (XSS)
 - o Command Injection
- Require authentication for protected endpoints.
- Apply API rate limiting to prevent abuse and denial-of-service attempts.
- Ensure error responses do not expose sensitive system details.

4. Data Protection & NDPR Compliance

- Collect only necessary personal data (data minimization).
- Obtain user consent for data collection and processing.
- Protect data through encryption:
 - o HTTPS/TLS for data in transit
 - o Database encryption where feasible
- Provide transparency through a privacy policy and user data rights.

5. Logging & Monitoring

- Maintain audit logs for key activities:
 - o Login attempts
 - o Orders and payments
 - o Listing updates
 - o Admin actions
- Protect logs from tampering and flag suspicious behavior.

6. Secure Development Practices

- No hardcoded secrets (API keys, DB credentials).
- Use environment variables for sensitive configurations.
- Keep dependencies updated and check for vulnerabilities.
- Follow OWASP secure coding standards.

7. Penetration Testing Readiness (Cyber Team Role)

- Conduct basic OWASP Top 10 security testing before deployment:
 - o Authentication testing
 - o Access control testing

- o Input validation testing

- o Vulnerability scanning